CSE 150: Introduction to Computer Networks

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Network Layer Part 1

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Strike – Wednesday and Thursday

- Lecture
  - Will be here on Thursday, don't anticipate any issues after 5pm
- Lab
  - Wednesday: see Canvas Announcement
  - Thursday: see Canvas Announcement
- Office Hours:
  - Lakshmi: see Canvas Announcement
  - Tyler: see Canvas Announcement

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Chapters 4 and 5: Reading Assignment

Chapter 4:

4.1 Overview of Network Layer

4.2 What's Inside a Router
- Limit reading to class discussion topics

4.3 IP: Internet Protocol
- datagram format
- IPv4 addressing
- NAT
- IPv6

Chapter 5:

5.1 Introduction

5.2 Routing Algorithms
- link state
- distance vector
- hierarchical routing

5.3 OSPF – skim

5.4 BGP

5.6 ICMP

Section numbers from 8th edition of text

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Final Project Groups

- Canvas->People->FinalProject
  - 2 per group
- If you have not heard from your partner, sign the sheet moving around the classroom
  - Email me!

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Final Project schedule

- Part 1: Sunday 11/24/2024 at 11:59pm NO LATE ASSIGNMENTS
- Part 2: Wednesday 12/4/2024 at 9pm; Demo Week 10
- Early Birds: Monday 11/27/2024 at 9pm
  Demo Monday 12/2/2024
  - Students done with their project by 11/27 can earn 15 points **extra credit**
  - Note: code and all components of project must be submitted by the above date to qualify
  - No late submissions for Early Bird extra credit.
- Demo
  - Required for credit
  - Signup sheet will be posted on Monday
- NO LATE ASSIGNMENTS ACCEPTED
  - Submit whatever you have finished by due date

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What's Left this quarter? (wish list)

Network:
- IP Addresses
- IP Forwarding
- Link State Routing
- Distance Vector Routing
- BGP: Border Gateway Protocol
- DHCP
- IP Fragmentation
- NAT Translation
- IPv6

Link Layer:
- Medium Access Control
  - CSMA protocols
- Ethernet
- WiFi
- ARP
- Packet Forwarding over the link
- Ethernet Switches

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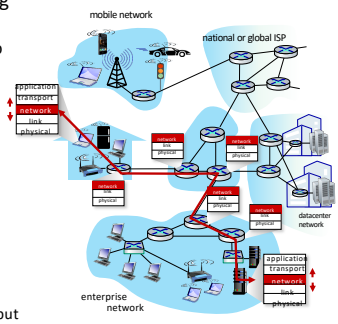
Network-layer services and protocols

- Transport segment from sending to receiving host
  - Sender: encapsulates segments into "datagrams", pass to link layer
  - Receiver: delivers segments to transport layer
- Network layer protocols in *every Internet device*: hosts, routers
- Routers:
  - L3 / IP datagrams
    - Move along end-end path
    - Every IP datagram gets its header examined ☺
    - Transferred from input port to output port



Two key network-layer functions

1. Forwarding: move packets from a router's input link to appropriate router output link

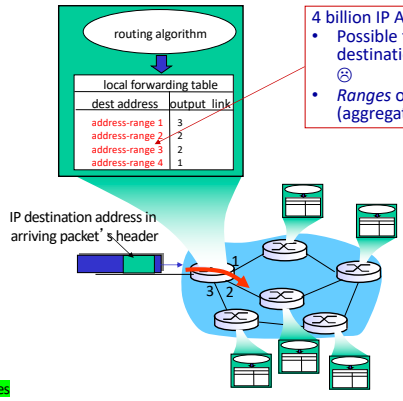
2. Routing: determine route taken by packets from source to destination
- *routing algorithms*

Analogy: taking a trip
- *Forwarding:* process of getting through single interchange
- *Routing:* process of planning trip from source to destination

forwarding

routing

Forwarding table

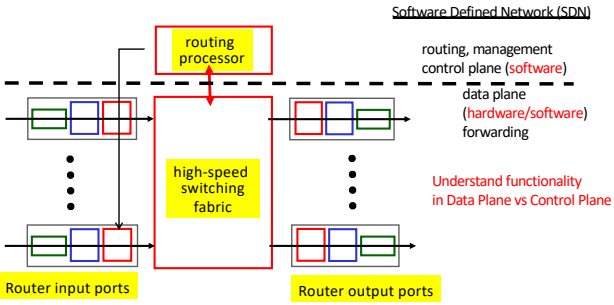


4 billion IP Addresses
- Possible to list individual destination addresses? ☹
- *Ranges* of addresses (aggregate table entries)

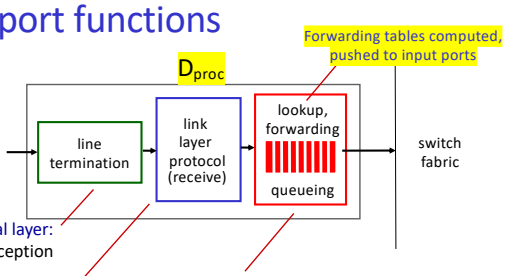
Smaller tables

Router Overview – Four Components

Two key functions:
- **Control Plane**: Run routing algorithms/protocol (RIP, OSPF, BGP)
- **Data Plane**: Forward packets from incoming to outgoing link (in to out ports)



Input port functions



Decentralized switching:
- Using header field values, lookup output port using forwarding table *("match plus action")*
- Traditional:
  - Destination-based forwarding: based on destination IP address
- (SDN) :
  - Generalized forwarding: based on any set of header field values

Forwarding by Address Range

32 Bit IP Address → Destination Address Range	Router Interface (output port)
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011111 11111111	2
Default (otherwise)	3

PRACTICE:
Write IP address ranges in dotted decimal format and mask (few more slides)
Line1 range: 200.23.16.0 … 200.23.23.255 → Network: 200.23.16.0 /21

Longest Prefix Matching

Longest prefix matching

When looking for match in forwarding table, choose ***longest*** Destination Address Range (prefix) that matches incoming datagram's destination address

Destination Address Range	Router Interface (output port)
11001000 00010111 00010*** *********	0
11001000 00010111 00011000 *********	1
11001000 00010111 00011*** *********	2
otherwise	3

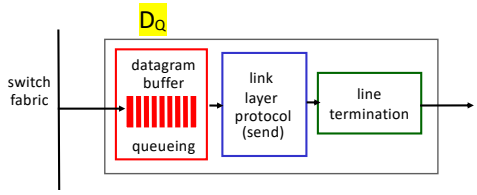
Examples:

Destination (binary): 11001000 00010111 00010110 10100001

Destination(binary): 11001000 00010111 00011000 10101010

out which interface?

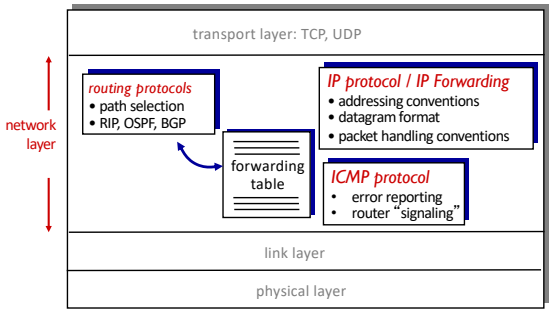
Output ports (after switch fabric)



- The QUEUE! "resting place" when arrival rate > departure rate
  - Scheduling discipline chooses among queued datagrams
- Delay and loss at output port
- (Note: can also occur at input port → all part of D_Q)

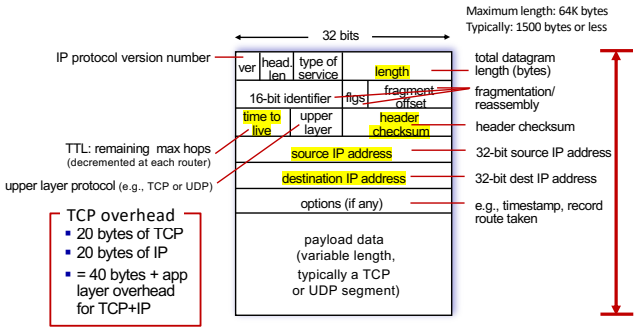
The network layer (AKA L3)

host, router network layer functions:



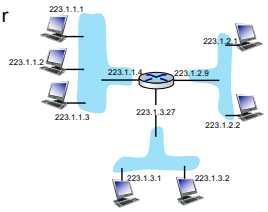
"IP traffic/datagram" is carried by the Network Layer

IP Header Format



IP addressing: introduction

- 32-bit IP address: identifier associated with each host or router *interface*
  - Network-wide identification
- interface: boundary between host/router and physical link
  - routers typically have multiple interfaces, each with IP Address
  - hosts typically have 1 or 2 interfaces
    - Your phone: (cellular connection + WiFi 802.11)



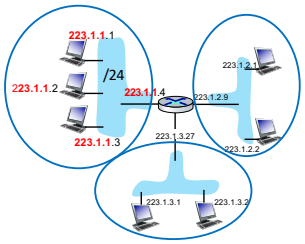
dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

What is a subnet ?

- IP address of all hosts on subnet have overlapping (same) bits
- Grouping of devices that can physically reach each other without router

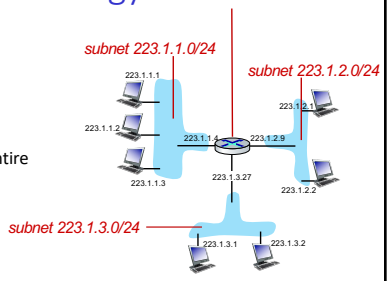


network consisting of 3 subnets

Subnets – it's a strategy

Benefits:

- Aggregate routes
  - Smaller forwarding table
- Router forwards packets to entire subnet over 1 interface



Shared part of IP address:
223.1.1, 223.1.2 and 223.1.3 are prefixes

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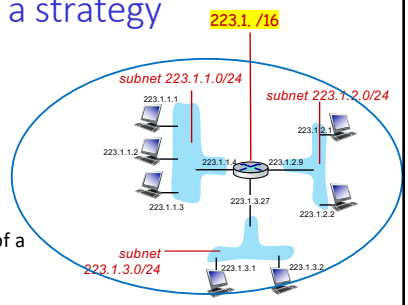
Subnets – it's a strategy

Benefits:

- keep internal network organization private
- Local organization

Each small subnet is part of a larger subnet



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Subnets – Local organization

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2 Ways to Divide IP Address Space

- CIDR: Classless Inter Domain Routing (current)
  - Subnets can be any size
  - Determined by variable length masks
- Classful Addresses (old)
  - A,B,C
  - Fixed size subnets/mask

Network/subnet	Host

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Classful Addresses

- Fixed size subnets
  - Class A
  - Class B
  - Class C

Subnet	Network Portion (Mask)	Host Portion	# IP Addresses
A	8	24	2^{24}
B	16	16	2^{16}
C	24	8	2^{8}

What if assigned network didn't need all of the addresses?

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IP addressing: CIDR

CIDR: Classless InterDomain Routing

- Mask identifies variable-length subnets
- No restriction on mask
- a.b.c.d/x, where /x is # bits in network portion

Network/subnet	Host

subnet part ← → host part

11001000 00010111 00010000 00000000

200.23.16.0 /24

/24 is the mask in CIDR notation
255.255.255.0 is the mask in dotted decimal notation

2 reserved addresses per subnet:
Network Address (all 0s in host portion)
Broadcast Address (all 1s in host portion)

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Masks

Network/subnet	Host

- Binary 1111 1111 1111 1111 1111 1111 0000 0000
- Dotted Decimal
- CIDR

Practice!

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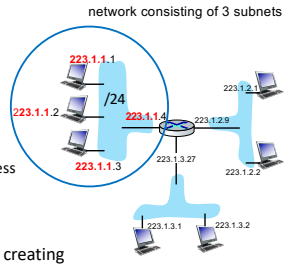
IP Address Hierarchy

2 parts of the IP address:

1) Subnet: high order bits in common
   - AKA "Prefix" or "Network Address"
2) Host: low order bits, different for each host

Mask /x defines the boundary:
- Number of bits in subnet portion of address
- Same mask for entire subnet



Recipe for identifying subnets:
- Visually: Detach each interface from router, creating "islands" of isolated networks (subnets)
- Same network address and mask

Network/subnet	Host

Homework:
1. Write Network Address (Zero out host bits)
2. Add 2 new hosts to subnet and assign addresses
3. How many hosts in each subnet?

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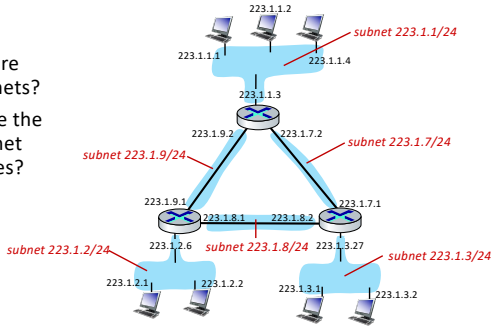
Homework: Subnets Practice

- Where are the subnets?
- What are the /24 subnet addresses?



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Example 1: Given IP Address and mask

Find the Network Address for the host with IP Address 200.23.16.54 /24

Method:
- bitwise AND or
- shortcut for easy masks

Mask /24 = 11111111 11111111 11111111 00000000

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Example 2: Assign Addresses to Hosts

200.23.16.0 /24

Network/subnet	Host
200.23.16	.0

IPs? # hosts?

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Example 3 Break it down - Make 16 Subnets 200.23.16.0 /24

Network/subnet	Host

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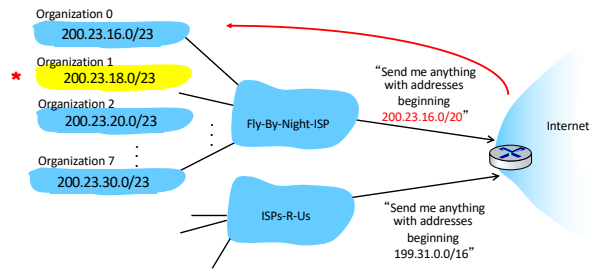
Fly-by-Night: route aggregation

Hierarchical addressing and reachability (in book)

Practice: 3 bits for the 8 subnets, write out the subnet prefixes



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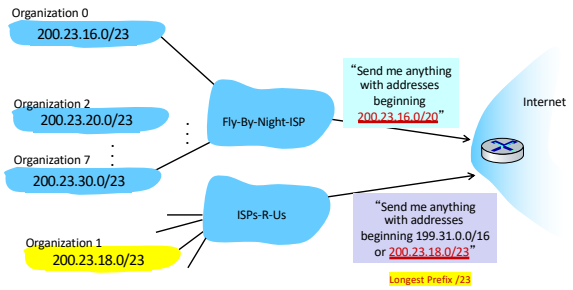
Hierarchical addressing: more specific routes

Organization 1 is sold to ISPs-R-Us!
ISPs-R-Us has a more specific route to Organization 1 (longer prefix)



Notice: 20.23.18.0 /23 is longer prefix than /20

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Why subnet?

- CIDR (Classless Inter-domain Routing) reduces waste of addresses compared to classful addressing
  - Understand why?
- Isolate traffic → single entry point
  - Security/privacy
  - Network resource utilization
    - Contention → throughput
- Allow organizational control over address space
  - Security/privacy
  - Organize networks
- Router Efficiency
  - Aggregation of routes reduces size of routing table
- Address Depletion Problem

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IP addresses for hosts

Q: How does a *host* get an IP address?

- Hard-coded by system admin in a file
  - Windows: control-panel->network->configuration->tcp/ip->properties
  - UNIX: /etc/rc.config
- DHCP: Dynamic Host Configuration Protocol: dynamically obtain address from as server
  - "plug-and-play"

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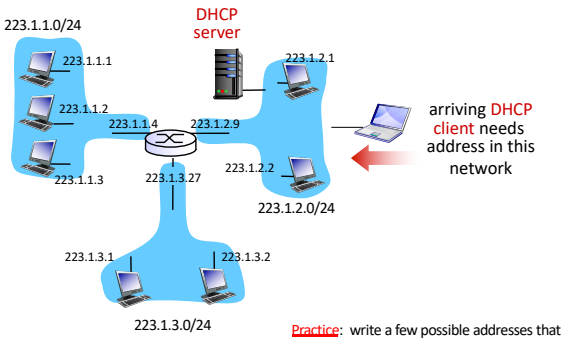
DHCP client-server scenario

2 ways: Server makes announcements or host broadcasts



Practice: write a few possible addresses that could be assigned to the arriving host

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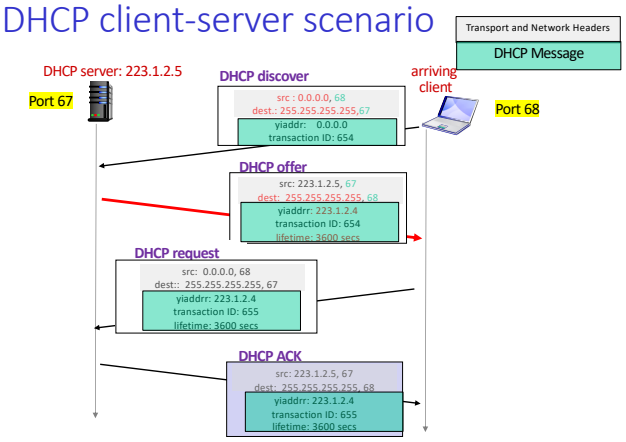
DHCP client-server scenario



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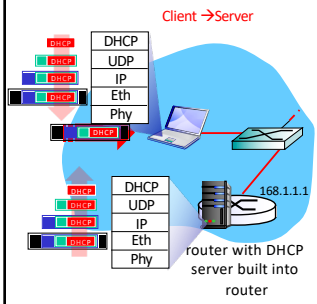
DHCP: Client Request – the headers



Good review of Encapsulation/Decapsulation

Client: DHCP request in UDP:

dst port 67, src port 68
→encapsulated in IP (broadcast)
→encapsulated in Ethernet broadcast frame (dest: FF:FF:FF:FF:FF:FF)

DHCP Server: request received

- Passed up protocol stack:
  - Ethernet →IP
  - →UDP
  - →DHCP server process

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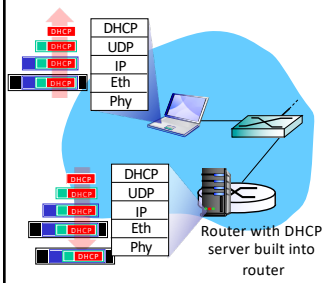
DHCP: Server ACK – the headers



Server→Client

DCP server: sends DHCP ACK with:

- IP address for Client
- First-hop router's IP address
- DNS server's name and IP address

DHCP ACK message forwarded to Client, de-mux up to DHCP Client process

Client:

- IP address and mask
- DNS server's name and IP
- First-hop router's IP

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DHCP: more than IP addresses

DHCP -- more than just IP address:

- First-hop router: IP Address
- DNS Server: Name and IP Address
- Network mask

All the tools needed to use the Internet!

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